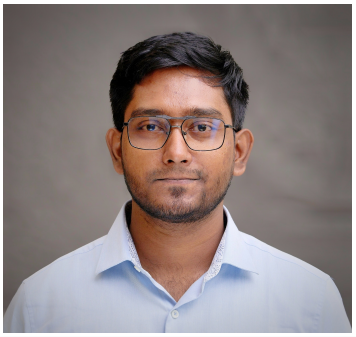
Mushfiqul Islam Chowdhury



AI / LLM Engineer | Open to full-time · Remote or hybrid

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Summary

AI engineer with **5+ years in software engineering**, including **3+ years building production LLM systems**: multi-agent (ReAct) orchestration, RAG pipelines, and high-throughput backend services in Python. Led Risk Sensor, a multilingual AI security-analysis platform that turned a one-week manual review into near-real-time automation. Now pursuing an M.Sc. in Artificial Intelligence at the University of Jyväskylä; IEEE-published in computer vision.

Professional Experience

Intern, Kimberton AI Studio *(credited as university professional training)* Jan 2026 – Apr 2026

- Developed the backend of a multi-model AI automation system, implementing stateful session management and model orchestration so users could choose and switch between multiple LLMs within the same workflow.
- Built the communication layer between users and models, maintaining per-user conversation state and routing each request to the user's selected model.

Software Engineer, Pipeline Bangladesh Ltd May 2022 – Dec 2025

- Architected and led Risk Sensor (risksensor.io), a production multilingual AI security-analysis platform using ReAct-style multi-agent orchestration and RAG, serving **10+ companies**.
- Replaced a manual process in which the security team delivered **1–2 reports per week** with **on-demand report generation**: every report is produced as soon as it is requested.
- Applied structured evaluation, reproducible runs, and fallback-aware design to keep LLM agent outputs reliable in production.
- Designed and optimized high-throughput data pipelines in Python processing **120K+ events per minute**, reducing memory consumption by **over 95%** while improving throughput and deployment efficiency.
- Owned services end to end, from requirements and design through code review, performance tuning, deployment, and operational handover; built internal web tools for AI-assisted review and monitoring.

Junior Software Engineer, Tirzok Private Limited Jun 2021 – Apr 2022

- Designed and implemented a graph-based reward engine using Neo4j and custom Odoo modules to automate rank calculation and reward distribution for e-commerce transactions.
- Served as the coordination point between backend engineering and product design to align implementation with product intent.

Junior Software Engineer (Part-time), Field Information Solutions Ltd Apr 2020 – May 2021
(Field Buzz)

- Redesigned role-based access control by modularizing permissions, improving usability and security.

Education

University of Jyväskylä, M.Sc. in Artificial Intelligence Aug 2025 – Present

- **GPA 4.4 / 5**; 86 of 120 ECTS completed in the first year.
- Relevant coursework: Deep Learning for Cognitive Computing (theory and developer tracks), Agent

Technologies for Developers, Natural Language Processing, Collective Intelligence and Agent Technology, Semantic Technologies for Developers, SOA and Cloud Computing, Data Visualization and Analysis.

- Focus: applied machine learning, NLP/LLM applications, and empirical evaluation of AI systems.

BRAC University, B.Sc. in Computer Science and Engineering Jan 2017 – Sep 2021

- Thesis: Real-Time Action Recognition from Video Footage (published at IEEE STI 2021).

Selected Projects

Picli (Co-founder & Backend Engineer, 2026) Python, FastAPI, PostgreSQL, Redis, AWS
picli.fi

- Co-founded Picli, a React Native (Expo) app that suggests one nearby event or small challenge each day; live on the App Store and Google Play.
- Built the FastAPI backend (PostgreSQL, Redis, ARQ, Firebase, S3), including the daily recommendation logic, and ran DevOps on AWS ECS with Cloudflare and Sentry, plus the store releases.

Sales Chatbot Python, LangChain, OpenAI API, PostgreSQL
github.com/mushfiqulIslam/sales-chatbot

- Built a generative AI chatbot for personalized product recommendations and order tracking using behavioral and country-level buying data.
- Designed a layered architecture separating AI orchestration, business logic, retrieval, and data access.

Publications

Real-Time Action Recognition from Video Footage 2021

Tasnim Sakib Apon, **Mushfiqul Islam Chowdhury**, Md Zubair Reza, et al. IEEE STI 2021 (conference publication). Built a 600-video CCTV dataset of violent actions and benchmarked pre-trained CNNs for real-time detection; VGG16 and MobileNetV2 performed best.
DOI: 10.1109/STI53101.2021.9732601 | arXiv: 2112.06456

Technical Skills

Languages: Python (primary, incl. asyncio), SQL, Go, JavaScript, TypeScript, Bash

LLM & Agents: RAG, ReAct multi-agent systems, LangChain/LangGraph, MCP, OpenAI API, Anthropic API, structured LLM evaluation

ML & Data Science: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas

Backend & Data: FastAPI, Django, REST APIs, SQLAlchemy, PostgreSQL, Redis, ARQ, Firebase, Kafka, Apache Spark, Qdrant, Neo4j, DynamoDB

Cloud & MLOps: AWS (EC2, ECS, S3, SageMaker, Bedrock), GCP (Vertex AI, Firestore), Cloudflare, Docker, Kubernetes, GitHub Actions (CI/CD), Sentry, MLflow, ZenML

Frontend & Tools: React, TDD (pytest), Claude Code, Cursor

Leadership & Community

Speaker, Google Developer Group (GDG) Dhaka: talk on "Career with AI" covering responsible and applied AI Sep 2024

General Secretary, BRAC University Computer Club: organized programming contests, workshops, and technical training Jul 2019 – Jul 2020

Languages

English (Fluent), Bengali (Native), Finnish (Basic; two university courses completed; actively studying)